SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I (***Sandesh S/192521078***) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: ***Sandesh S***

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

ANSWER 1 – Type-1 vs Type-2 Hypervisors

For a cloud datacenter hosting mission-critical applications, a Type-1 hypervisor is the better choice.

Type-1 hypervisor:

- Runs directly on the physical hardware without a host operating system.
- Provides better performance because there is less software overhead between the VM and hardware.
- Gives stronger isolation and a smaller attack surface than a hosted hypervisor.
- Supports centralized management, resource allocation, monitoring, and high availability in enterprise datacenters.

Type-2 hypervisor:

- Runs on top of a host operating system.
- Is easier to install and is useful for testing, development, and desktop virtualization.
- Has additional overhead because requests pass through the host operating system.
- Security depends partly on the security of the host operating system, increasing the possible attack surface.

Analysis:

For mission-critical workloads, high CPU and memory efficiency, predictable performance, strong isolation, and centralized management are important. Type-1 hypervisors provide these advantages and are therefore more suitable for production cloud datacenters.

Justification:

A Type-1 hypervisor should be selected because it provides better performance, stronger security isolation, and better manageability for critical cloud applications.

ANSWER 2 – Workload Isolation and Attack Surfaces

Virtualization allows finance, healthcare, and student workloads to share the same physical hardware while remaining logically separated. Each workload can run inside its own virtual machine with allocated CPU, memory, storage, and virtual networking resources.

Isolation:

• The hypervisor controls access between virtual machines and prevents one VM from directly accessing another VM's memory or virtual hardware.

• Separate virtual networks, VLANs or security groups can restrict communication between workloads.

• Resource allocation and access controls help prevent one workload from consuming all available resources.

• Storage and identity permissions can be configured so that sensitive data is accessible only to authorized workloads.

Two major attack surfaces are:

1. Hypervisor attack surface – A vulnerability in the hypervisor or its management interface could allow an attacker to affect multiple VMs.

2. Virtual network/VM attack surface – Weak network controls, vulnerable guest operating systems, or insecure virtual interfaces can allow attacks between workloads or from external systems.

Security measures such as patching the hypervisor, strong authentication, network segmentation, least-privilege access, monitoring, and encryption reduce these risks.

ANSWER 3 – VM Host Utilization Analysis

The utilization snapshot shows that the VM host is under heavy resource pressure.

Given values:

• CPU utilization = 92%

• Memory utilization = 88%

• Storage I/O latency = 35 ms

• Average VM boot time = 170 seconds

Likely bottlenecks:

1. CPU: At 92%, the host has very little CPU capacity available for workload spikes. CPU contention can increase VM response time.

2. Memory: At 88%, available memory is limited. Memory pressure may cause swapping or increased contention, reducing performance.

3. Storage I/O: A latency of 35 ms indicates slow storage operations. VM disk access and boot operations can therefore take longer.

4. VM boot time: An average boot time of 170 seconds is high and is consistent with CPU, memory, and storage contention.

Corrective actions:

• Add CPU capacity or migrate some VMs to another host.

• Increase available memory and rebalance memory-heavy VMs.

• Move workloads to faster storage and optimize storage I/O.

• Reduce unnecessary VM resource consumption and schedule heavy workloads appropriately.

• Use workload balancing or live migration to distribute VMs across hosts.

The main problem is resource contention across CPU, memory, and storage. Balancing the workload and increasing capacity where required should improve VM boot time and overall performance.

ANSWER 4 – Software Defined Datacenter Architecture

A Software Defined Datacenter (SDDC) improves agility by using virtualization and software-based control to manage compute, storage, and networking resources. In a traditional datacenter, these resources are more dependent on manually configured physical hardware.

Compute virtualization:

- Physical servers can run multiple VMs.
- VMs can be created, resized, migrated, or removed quickly.
- Resources can be allocated according to workload requirements.

Storage virtualization:

- Physical storage from different systems can be presented as shared logical storage.
- Storage can be provisioned and reallocated through software.
- This reduces the need for manual physical configuration.

Network virtualization:

- Virtual networks can be created and modified without changing physical cabling.
- Software-based policies can control connectivity and security.
- Network segmentation can be created quickly for different applications.

Agility improvement:

SDDC enables automation, centralized management, rapid provisioning, easier scaling, and faster deployment of applications. A traditional datacenter generally requires more manual configuration of servers, storage devices, and network equipment.

Therefore, SDDC provides greater flexibility and faster response to changing business and application requirements while improving resource utilization.

ANSWER 5 – Secure Identity and Privacy Design for Multi-Cloud Federation

The federation project fails because different cloud providers may use different identity formats, authentication methods, roles, attributes, and access-control models. If these identities are not mapped correctly, users may be rejected or receive incorrect permissions.

Identity design:

- Use a trusted identity provider (IdP) to provide a common source of identity.
- Use federation standards such as SAML or OpenID Connect to allow secure authentication across providers.
- Map user identities and roles from the common identity system to the roles required by each cloud provider.
- Apply single sign-on with multi-factor authentication for stronger security.
- Use least-privilege access so users receive only the permissions required for their work.
- Regularly review and revoke unnecessary access.

Privacy design:

- Share only the minimum identity attributes required by each provider.
- Encrypt identity and sensitive data during transmission and storage.
- Use strong authentication and secure tokens with limited lifetime.
- Maintain audit logs to detect unauthorized access and support accountability.
- Establish clear policies for data location, retention, and access between cloud providers.

A secure federation architecture therefore requires common identity standards, correct identity-to-role mapping, strong authentication, minimum attribute sharing, encryption, and continuous monitoring. This can resolve identity mismatches while preserving security and privacy across the multi-cloud environment.